



SAMUEL ARTA RAJU MANIK

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Medan, Sumatera Utara, Indonesia

Aspiring Fullstack Developer with internship experience in enterprise web application development and software solutions. Experienced in designing and developing responsive web applications, internal dashboard systems, and AI-powered chatbot solutions using Next.js, React.js, TypeScript, JavaScript, MySQL, PostgreSQL, LangChain, and OpenAI APIs. Skilled in frontend and backend development, database management, system integration, and problem-solving through academic and professional projects. Passionate about modern web technologies, software architecture, and continuous learning, while currently seeking a full-time opportunity to further develop expertise and contribute to a dynamic technology organization.

Education Level

Institut Teknologi Del - Laguboti, North Sumatra, Indonesia

Aug 2023 - Sep 2026 (Expected)

Associate Degree in Computer Engineering Technology

- Relevant Coursework: Software Engineering, Database Systems, Web Development, Artificial Intelligence, Data Structures & Algorithms, Computer Networks, and Internet of Things (IoT).
- Developed academic and industry projects in web development, artificial intelligence, and IoT, including RAG-based chatbot systems, enterprise web applications, and automation solutions.
- Applied modern technologies such as React.js, Next.js, TypeScript, MySQL, PostgreSQL, LangChain, and OpenAI APIs while actively participating in student organizations and campus events to strengthen leadership, communication, and teamwork skills.

Professional Experience

PT. Pelindo Multi Terminal - Medan Belawan, Indonesia

Feb 2026 - Present

Fullstack Developer (Internship)

PT Pelindo Multi Terminal is a state-owned port and logistics company specializing in terminal operations, cargo handling, and integrated logistics services across Indonesia. The company supports national trade and supply chain activities through efficient port management and logistics operations. As part of its digital transformation initiatives, Pelindo Multi Terminal leverages technology-driven solutions to enhance operational efficiency, workforce management, and business performance. The company continuously develops digital systems to support data-driven decision-making and process optimization across its operational network.

- Developed and maintained an internal HR dashboard system to manage and visualize workforce data across multiple entities, improving data accessibility and reporting efficiency.
- Implemented frontend and backend features using Next.js, React.js, TypeScript, JavaScript, and MySQL to support workforce management processes.
- Collaborated with stakeholders to enhance data processing workflows and support data-driven decision-making through dashboard visualization and reporting functionalities. Contributed to system maintenance, feature enhancements, and application performance optimization within an enterprise environment.

Projects

Fine-Tuning a Language Model for Adaptive Computer Networks Learning Tasks

Aug 2025 - Jan 2026

Institut Teknologi Del

An undergraduate final project conducted at Institut Teknologi Del focused on developing and fine-tuning a Large Language Model (LLM) to support adaptive learning in computer networking education through AI-powered and context-aware learning assistance.

- Developed an AI-powered learning assistant utilizing Retrieval-Augmented Generation (RAG) and Large Language Models to support adaptive learning within computer networking topics.
- Designed retrieval pipelines and contextual response generation mechanisms to improve learning relevance and user engagement.
- Technologies: JavaScript, LangChain, OpenAI API, Supabase

Implementing RAG for Analogy-Based Chatbot Responses

Feb 2025 - Jun 2025

Institut Teknologi Del

An academic software engineering project conducted at Institut Teknologi Del focused on developing a Retrieval-Augmented Generation (RAG) chatbot capable of generating analogy-based responses to improve contextual understanding and communication effectiveness.

- Built a Retrieval-Augmented Generation (RAG) chatbot capable of generating analogy-based responses to improve contextual understanding and communication effectiveness.
- Integrated vector-based retrieval mechanisms and language models to enhance response relevance and reasoning accuracy.
- Developed backend workflows for document retrieval, embedding generation, and conversational response processing.
- Technologies: JavaScript, Node.js, Express.js, LangChain, OpenAI API, Supabase (PostgreSQL), HTML/CSS, Git, GitHub.

An academic web development project conducted at Institut Teknologi Del focused on designing and developing an interactive website that facilitates parent–child communication through analogy-based explanations and conversational learning approaches.

- Designed and developed an interactive web application that facilitates parent–child communication through analogy-based explanations and conversational learning approaches.
- Implemented responsive user interfaces and database integration to support content management and user interaction.
- Improved accessibility and engagement by presenting complex concepts through relatable analogy-driven communication methods.
- Technologies: HTML, CSS, JavaScript, MySQL

An Internet of Things (IoT) project conducted at Institut Teknologi Del focused on developing a smart door system using sensors and microcontrollers to provide automated access control and monitoring capabilities.

- Developed an IoT-based smart door system utilizing sensors and microcontrollers to automate access control and monitoring processes.
- Integrated hardware and software components to improve security, reliability, and real-time system responsiveness.
- Conducted testing and validation to ensure stable operation under various usage scenarios.

Skill

- **Soft Skills:** Problem Solving, Team Collaboration, Communication Leadership, Adaptability, Time Management, Public Speaking, Active Learning, Analytical Thinking
- **Hard Skills:** JavaScript, TypeScript, PHP, React.js, Next.js, Node.js, HTML5, CSS3, REST API Development, MySQL, PostgreSQL, Supabase, LangChain, Retrieval-Augmented Generation (RAG), OpenAI API, Git, GitHub
- **Achievements** (2025): Huawei ICT Competition 2025–2026 National Level – Network Track
- **Webinars Attended** (2025): Website Development: Back End